

Revision session 1

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$$|l_A, m_A; l_B, m_B\rangle \rightarrow |J, m_J\rangle \quad (1)$$

where m_J is the z component. We had the rules that. In question $l_A = 2$ and $l_B = 1$

$$J \in \{|l_A - l_B|, \dots, l_A + l_B\} \in \{1, 2, 3\} \quad (2)$$

We have the cases WE MUST SPELL OUT FOR FULL MARKS

$$m_{J=1} \in \{-1, 0, 1\} \quad (3)$$

$$m_{J=2} \in \{-2, -1, 0, 1, 2\} \quad (4)$$

$$m_{J=3} \in \{-3, -2, -1, 0, 1, 2, 3\} \quad (5)$$

Also we require

$$m_J = m_A + m_B \quad (6)$$

(ii)

$$\hat{L}_\pm |3, 3\rangle = \hbar \sqrt{l(l+1) - m(m \pm 1)} |l, m \pm 1\rangle \quad (7)$$

$$|3, 3\rangle = |2, 2; 1, 1\rangle \quad (8)$$

$$\hat{L}_- |3, 3\rangle = \hat{L}_- |2, 2; 1, 1\rangle \quad (9)$$

$$\sqrt{6}\hbar |3, 2\rangle = 2\hbar |2, 1; 1, 1\rangle + \sqrt{2}\hbar |2, 2; 1, 0\rangle \quad (10)$$

Sum of the squares of the coef on RHS should be square of coef on LHS

$$|3, 2\rangle = \sqrt{\frac{2}{3}} |2, 1; 1, 1\rangle + \sqrt{\frac{1}{3}} |2, 2; 1, 0\rangle \quad (11)$$

$$(12)$$

Check sum of squares equals 1.

$$\sqrt{10}\hbar |3, 1\rangle = \sqrt{\frac{2}{3}} \sqrt{6}\hbar |2, 0; 1, 1\rangle + \sqrt{\frac{2}{3}} \sqrt{2}\hbar |2, 1; 1, 0\rangle \quad (13)$$

We get

$$|3, 1\rangle = \sqrt{\frac{2}{5}} |2, 0; 1, 1\rangle + \sqrt{\frac{8}{15}} |2, 1; 1, 0\rangle + \sqrt{\frac{1}{15}} |2, 2; 1, -1\rangle \quad (14)$$

(iii)

$$P(m_A = 1 | J = 3, m_J = 1) = \left(\sqrt{\frac{8}{15}} \right)^2 \quad (15)$$

(iv)

we have the set with $m_J = 2$

$$\{|3, 2\rangle, |2, 2\rangle\} \leftrightarrow \{|2, 1; 1, 1\rangle, |2, 2; 1, 0\rangle\} \quad (16)$$

$$\text{impose } \langle 3, 2|2, 2\rangle \text{ to find } |2, 2\rangle \quad (17)$$

$$|2, 2\rangle = \alpha|2, 1; 1, 1\rangle + \beta|2, 2; 1, 0\rangle \text{ with } |\alpha|^2 + |\beta|^2 = 1 \quad (18)$$

$$0 = \sqrt{\frac{2}{3}}\alpha + \sqrt{\frac{1}{3}}\beta \quad (19)$$

$$\Rightarrow \alpha = \sqrt{\frac{1}{3}}, \beta = -\sqrt{\frac{2}{3}} \quad (20)$$

sum of squares on both sides should match, sum of m_J should match

(Part B)

$$E_{var} = \langle \psi | \hat{H} | \psi \rangle \quad (21)$$

$$E_{var} \leq E_0 \quad (22)$$

$$(23)$$

Where the EV turns into integration

$$\int \psi^*(x) (\hat{H}\psi(x)) dx_1 dx_2 \quad (24)$$

Normalized the guess $\langle \psi | \psi \rangle = 1$

We may have a free parameter. We know there is an upper bound so we can minimise the variational parameter. let derivative equal 0.

Question L2

0.0.1 (1 part b)

$$\mathcal{H} = -\frac{\hbar^2}{2m_p} \frac{\partial^2}{\partial x_p^2} - \frac{\hbar^2}{2m_Q} \frac{\partial^2}{\partial x_Q^2} - v_0 \delta(x_p - x_Q) \quad (25)$$

Variational method

$$\psi(x_p, x_Q) = \frac{2}{a} \sin\left(\frac{\pi x_p}{a}\right) \sin\left(\frac{\pi x_Q}{a}\right) \quad (26)$$

$$\text{check } \int \int |\psi_{guess}|^2 dx_p dx_Q = 1 \quad (27)$$

$$E_{var} = \langle \psi | \mathcal{H} | \psi \rangle \quad (28)$$

$$E_{var} = \left(\frac{2}{a}\right)^2 \int_0^a \int_0^a \left(\sin\left(\frac{\pi x_p}{a}\right) \sin\left(\frac{\pi x_Q}{a}\right)\right) \mathcal{H} \left(\sin\left(\frac{\pi x_p}{a}\right) \sin\left(\frac{\pi x_Q}{a}\right)\right) dx_p dx_Q \quad (29)$$

$$\mathcal{H} \left(\sin\left(\frac{\pi x_p}{a}\right) \sin\left(\frac{\pi x_Q}{a}\right)\right) = \left[\frac{\hbar^2 \pi^2}{2ma^2} + \frac{\hbar^2 \pi^2}{2m_p a^2} + \frac{\hbar^2 \pi^2}{2m_p a^2} \right] \quad (30)$$

$$E_{var} = \frac{\hbar^2 \pi^2}{2m_p a^2} + \frac{\hbar^2 \pi^2}{2m_Q a^2} - v_0 \left(\frac{2}{a}\right)^2 \frac{3a}{8} \quad (31)$$

$$-\frac{3v_0}{2a} \quad (32)$$

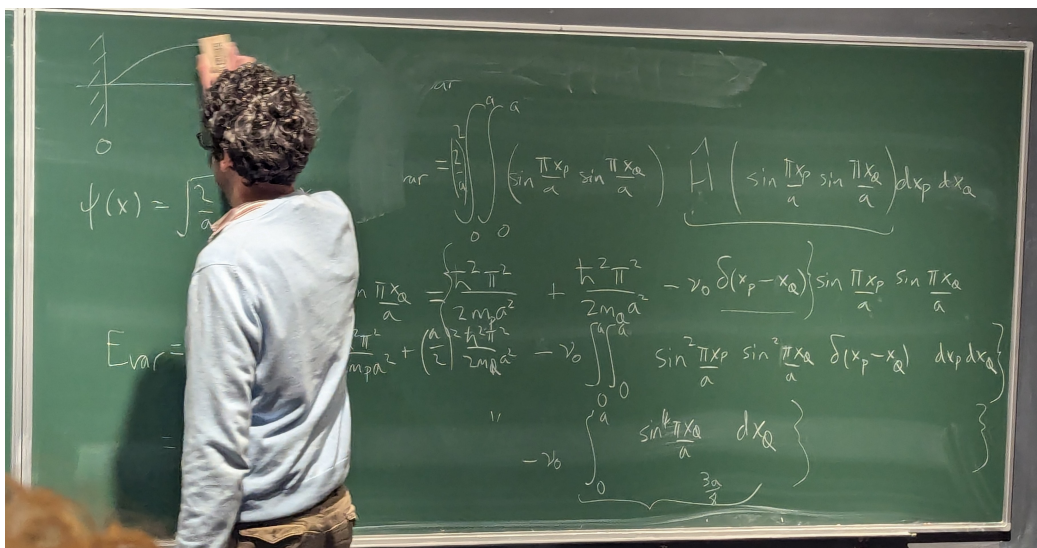


Figure 1: 1.1.jpg

Question 2

0.0.2 (a)

$$(\mathcal{H}^0 + gV) (|\psi_k^0\rangle + g|\psi_k^1\rangle) = (E_k^0 + gE_k^1) (|\psi_k^0\rangle + g|\psi_k^1\rangle) \quad (33)$$

$$g(V|\psi_k^0\rangle + \mathcal{H}^0|\psi_k^1\rangle) = g(E_k^1|\psi_k^0\rangle + E_k^0|\psi_k^1\rangle) \quad (34)$$

LHS we have a first order zeroth order and vice versa, same for RHS, good check

Group order of terms

$$(\mathcal{H}^0 - E_k^0) |\psi_k^1\rangle = (E_k^1 - V) |\psi_k^0\rangle \quad (35)$$

'Energy shift $\rightarrow \langle \psi_k^0 | \dots \rangle$ '

$$\langle \psi_k^0 | (\mathcal{H}^0 - E_k^0) |\psi_k^1\rangle = E_k^1 - \langle \psi_k^0 | V | \psi_k^0 \rangle = 0 \quad (36)$$

$$\implies \langle \psi_k^0 | V | \psi_k^0 \rangle = E_k^1 \quad (37)$$

$$|\psi_k^1\rangle = \sum_{l \neq k} c_{lk} |\psi_l^0\rangle \quad (38)$$

L3

Two series in small parameter, g

$$E_k = E_k^0 + gE_k^1 + g^2E_k^2 + \dots \quad (39)$$

$$|\psi_k\rangle = |\psi_k^0\rangle + g|\psi_k^1\rangle \quad (40)$$

As we saw last lecture

$$(\mathcal{H}^0 - E_k^0) |\psi_k^1\rangle = (E_k^1 - V) |\psi_k^0\rangle \quad (41)$$

apply energy shift $\langle \psi_j^0 | * **$ and we have $E_k^1 = \langle \psi_k^0 | V | \psi_k^0 \rangle$

$$|\psi_k^1\rangle = \sum_{l \neq k} c_{lk} |\psi_l^0\rangle \quad (42)$$

$j \neq k$

$$c_{jk} (E_j^0 - E_k^0) = \langle \psi_j^0 | \hat{V} | \psi_k^0 \rangle \quad (43)$$

$$|\psi_k^1\rangle = \sum_{j \neq k} \frac{\langle \psi_j^0 | V | \psi_k^0 \rangle}{E_j^0 - E_k^0} |\psi_j^0\rangle \quad (44)$$

we have the potential

$$g\hat{V} = gx(a-x) \quad (45)$$

the ground state is given by $k = 1$

$$E_k^1 = \langle \psi_k^0 | \hat{V} | \psi_k^0 \rangle \quad (46)$$

$$= \int \left(\sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \right) x(x-a) \left(\sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \right) dx \quad (47)$$

$$= a^2 \left(\frac{1}{6} + \frac{1}{2\pi^2} \right) \quad (48)$$

the first excited state $k = 2$

$$E_2^1 = a^2 \left(\frac{1}{6} + \frac{1}{8\pi^2} \right) \quad (49)$$

(C)

$$\hat{H}|1\rangle = 3|1\rangle \quad (50)$$

$$E = 1 \implies |\psi_1\rangle = \frac{1}{\sqrt{2}} (|2\rangle + |3\rangle) \quad (51)$$

$$E = 3 \implies |\psi_3\rangle = \frac{1}{\sqrt{2}} (|2\rangle - |3\rangle) \quad (52)$$

$$\hat{V}|1\rangle = |3\rangle \quad (53)$$

$$\hat{V}|2\rangle = 0 \quad (54)$$

$$\hat{V}|3\rangle = |1\rangle \quad (55)$$

$$\hat{V} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad (56)$$

The linear algebra way

$$E_1^1 = \frac{1}{\sqrt{2}} (0, 1, 1) \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 \quad (57)$$

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 \end{bmatrix} \quad (58)$$

$$E_{2,3}^1 = \pm \frac{1}{\sqrt{2}} \quad (59)$$